Assignment 2: Quantum algorithms
COMP3366: Quantum Algorithms and Computer Architecture

Release date: October 6, 2026

Due date: November 12, 2026

Learning Outcomes:

- (O1) Concepts
- (O2) Problem solving
- (O3) Quantum algorithm design

Note: When solving the exercises, you can use the content of the lectures (e.g., a state is entangled if it is not a product state) directly.

Question 1 (10 Points): Quantum Fourier basis (O1, O2)

For $d \in \mathbb{N}$ and any $\omega \in \mathbb{C}$ with $|\omega| = 1$, define the vectors:

$$|\psi_k\rangle := \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \omega^{k \cdot j} |j\rangle \qquad k = 0, 1, ..., d-1$$

on a d-dimensional linear space.
a) Compute the inner product $\langle \psi_k | \psi_l \rangle$ for arbitrary k, l. (2 points)

b) Find ALL ω such that $\{|\psi_k\rangle\}_{k=0}^{d-1}$ is an orthonormal basis. (3 points)

c) Suppose Alice wants to generate the Fourier basis $\{|\psi_k\rangle\}_{k=0}^{d-1}$ with $\omega = e^{2\pi i/d}$, but accidentally generated $\{|\psi_k\rangle\}_{k=0}^{d-1}$ with $\omega = e^{2\pi i m/d}$ for some integer m. For $d = 8$ (i.e., 3 qubits), find all m such that Alice still achieves her goal of generating $\{|\psi_k\rangle\}_{k=0}^{d-1}$ with $\omega = e^{2\pi i/d}$ such that each $|\psi_k\rangle$ matches exactly the target $|\psi_k\rangle$ with the same index k. (5 points)

Question 2 (20 Points): Quantum Fourier Transform (QFT) (O2, O3)

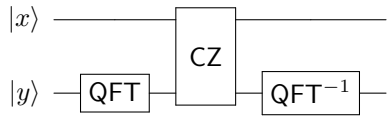
Recall that the n-qubit Quantum Fourier Transform (QFT) applies the following:

$$\mathsf{QFT}(|x\rangle) = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} \exp\left(i \cdot \frac{2\pi x k}{2^n}\right) |k\rangle .$$

a) Prove that the QFT operation is unitary. (4 points)
b) Is QFT self-inverse (i.e., $\mathsf{QFT} = \mathsf{QFT}^{-1}$)? Justify your answer. (4 points)
c) Given two computational basis $|x\rangle$, $|y\rangle$, we define the generalized $2n$-qubit control-Z gate, via

$$\mathsf{CZ}\,|x\rangle\,|y\rangle = e^{i \cdot \frac{2\pi x y}{2^n}}\,|x\rangle\,|y\rangle\,, \quad \forall x, y \in \{0,1\}^n.$$

Derive the computational result of the following operation:



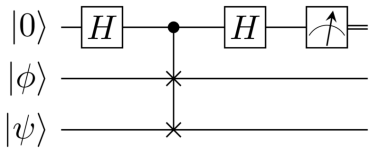
(6 points)
d) For $n = 2$, draw a quantum circuit consisting of CNOT gates and single-qubit gates that implements the following ADDITION operation:

$$|x\rangle\,|y\rangle \mapsto |x\rangle\,|x+y \mod 2^n\rangle\,.$$

(6 points)

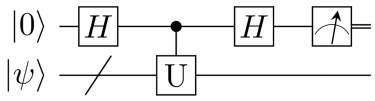
Question 3 (25 Points): the SWAP test and the Hadamard test (O2,O3)

The following circuit is a very useful gadget in quantum algorithms:



Here the 3-qubit gate is a controlled-SWAP gate ($|0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes SWAP$), and $|\psi\rangle, |\phi\rangle$ are generic qubit states.
[Hint: Recall the Born rule for partial measurement on a part of a two-register $|\Psi\rangle$: $P(x) = \langle\Psi|(|x\rangle\langle x| \otimes I)|\Psi\rangle$, where I is the identity on the part that is not measured.]
a) What is the probability of getting outcome 0 in the measurement performed on the first qubit? (5 points)
b) Which figure of merit (about a pair of quantum states) can we evaluate using this circuit? (5 points)
c) What is the probability of getting outcome 0, if we replace $|\phi\rangle$ and $|\psi\rangle$ by the Bell state $(1/\sqrt{2})(|01\rangle - |10\rangle$? (5 points)

Consider the following circuit:



Here the $|\psi\rangle$ is a generic quantum state.
d) What is the probability of getting outcome 0 in the measurement performed on the first qubit? (5 points)
e) What is the relation between this circuit and the quantum phase estimation algorithm? (5 points)

Question 4 (20 Points): Shor (O2,O3)

Review the slides on factorization and order-finding. Shor's algorithm on factorization consists of a quantum subroutine of order-finding and other classical subroutines.
a) Write down explicitly Shor's algorithm for the factorization of $N = 33$. (10 points).
[Hint: refer to the slides for the case of $N = 15$.]
b) Describe the snapshots of the quantum state of the order-finding subroutine when $x = 2$ (5 points).

c) Consider applying Shor's algorithm to $N = 27$ and discuss potential issues, if any. (5 points)

Question 5 (25 Points): Qiskit Algorithm Design: Coding with Grover Search (O3)

Consider Grover's search algorithm on a search space of $N = 8$ computational basis states represented by three qubits. Suppose the unique marked state is $|101\rangle$.

a) In Qiskit, construct a three-qubit marked phase oracle O such that

$$O|x\rangle = \begin{cases} -|x\rangle, & x = 101, \\ |x\rangle, & x \neq 101. \end{cases}$$

Provide your Qiskit code and draw the corresponding circuit here. (6 points)

b) In Qiskit, construct the following Grover diffusion operator

$$D = 2|s\rangle\langle s| - I, |s\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{7} |x\rangle .$$

Combine the oracle and diffusion operator to implement two Grover iterations, starting from the all-0 state. Provide your Qiskit code and draw the complete circuit here. (8 points)

c) Measure all three qubits after the two Grover iterations and run the circuit on a simulator with at least 2048 shots (a simulator function is included in our tutorial handout). Report the measurement counts or histogram and the estimated probability of obtaining the marked state $|101\rangle$. (6 points)

d) For one marked state among $N = 8$ states, let

$$\sin^2 \theta = \frac{1}{8}.$$

Compute the theoretical probability of obtaining the marked state after two Grover iterations and compare it with your simulation result in **(c)**. (5 points)